

Background-Matched MALDI-AMR Audit

Rough reviewer-facing draft explaining the tables, figures, and argument

Central Thesis

This project is no longer just a MALDI-TOF AMR model. It is a background-aware evaluation framework for asking whether apparent resistance prediction reflects focal-drug signal or resistant-population background.

The core claim is that raw external AUC can be biologically ambiguous. A model may appear to predict resistance while using spectral correlates of lineage, co-resistance, or hospital-specific resistant ecology. Background-matched evaluation provides a practical audit of that ambiguity using spectra, AST labels, isolate IDs, and model prediction scores.

- Layer 1: external transfer audit establishes that generalization is pair-dependent.
- Layer 2: background-matched contrastive audit is the core proof; it asks whether focal-drug prediction survives co-resistance matching.
- Layer 3: cross-resistance network makes the exploitable label ecology visible.
- Layer 4: public WGS/proteomic cross-reference grounds the interpretation in lineage biology.

Figure 5. Framework schematic

Background-Matched MALDI-AMR Audit

A model-agnostic framework for separating focal resistance signal from resistant-population background.



What it measures. This figure is conceptual rather than statistical. It lays out the four-layer audit pipeline: external transfer, background matching, cross-resistance ecology, and public WGS/proteomic cross-reference.

What a reviewer should see. The paper is not arguing from a single AUC table. It is building a linked framework: first identify raw external performance, then ask whether the signal survives background control, then explain why background is exploitable, then support the biology with WGS-linked public data.

How it supports the paper. It positions the contribution as an evaluation framework, not a model architecture claim. This matters because the method can be applied to CNNs, LGBM, Weis-style models, or any future MALDI-AMR predictor that outputs isolate-level prediction scores.

Caveat. The schematic does not prove the biology by itself. It should be used as the orienting figure or graphical abstract, not as empirical evidence.

Table 1. Primary background-matched transfer audit

What it measures. For each organism-drug/site comparison, this table reports raw external AUC, matched AUC, and stratum-centered AUC after grouping isolates by co-resistance background.

Key result. Ciprofloxacin retains more signal after matching than Amox-Clav, while several cephalosporin rows are explicitly flagged as low-retention rather than overinterpreted.

How it supports the paper. This is the primary evidence that raw MALDI-AMR performance needs a background-aware interpretation. It transforms AUC from a single performance number into a diagnostic: how much signal remains after background matching?

Reviewer caveat. Rows with low matched retention or few valid strata are not decisive. The table handles this by labeling adequacy, which is exactly the right conservative move.

evidence_tier	pair	site	raw_auc_95ci	stratum_cente red_auc_95ci	raw_to_center ed_delta	matched_rete ntion_pct	adequacy	interpretation
Main contrast	E. coli / Cipro	A-2018	0.823 (0.799-0.849)	0.703 (0.664-0.747)	0.120	61.900	interpretable	Survives background matching
Cautionary support	E. coli / Cipro	DRIAMS-B	0.774 (0.692-0.841)	0.860 (0.636-1.000)	-0.086	11.800	caution_low_n _matched	Suggestive residual signal, low support
Main contrast	E. coli / Cipro	DRIAMS-C	0.750 (0.702-0.795)	0.646 (0.576-0.719)	0.104	47.600	interpretable	Survives background matching
Main contrast	E. coli / Cipro	DRIAMS-D	0.671 (0.640-0.700)	0.596 (0.565-0.624)	0.075	98.400	interpretable	Survives background matching
Not main evidence	E. coli / Norfloxacin	A-2018	0.743 (0.678-0.803)			0.000	not_interpreta ble_no_valid_ strata	Not interpretable after matching
Not main evidence	E. coli / Norfloxacin	DRIAMS-B	0.703 (0.631-0.783)	0.343 (0.176-0.578)	0.360	54.300	interpretable	Near chance after matching
Not main evidence	E. coli / Norfloxacin	DRIAMS-C	0.779 (0.727-0.835)			0.000	not_interpreta ble_no_valid_ strata	Not interpretable after matching
Main contrast	E. coli / Amox-Clav	A-2018	0.650 (0.619-0.685)	0.541 (0.504-0.576)	0.109	92.200	interpretable	Survives background matching
Main contrast	E. coli / Amox-Clav	DRIAMS-B	0.694 (0.611-0.765)	0.576 (0.410-0.715)	0.118	62.100	interpretable	Weak or uncertain residual signal
Main contrast	E. coli / Amox-Clav	DRIAMS-C	0.535 (0.494-0.577)	0.497 (0.449-0.547)	0.038	89.200	interpretable	Collapses to chance after matching
Main contrast	E. coli / Amox-Clav	DRIAMS-D	0.557 (0.521-0.590)	0.486 (0.450-0.521)	0.070	97.400	interpretable	Collapses to chance after matching
Cephalosporin support	E. coli / Ceftriaxone	A-2018	0.863 (0.835-0.889)	0.645 (0.520-0.762)	0.218	10.900	interpretable	Survives, but strongly attenuated

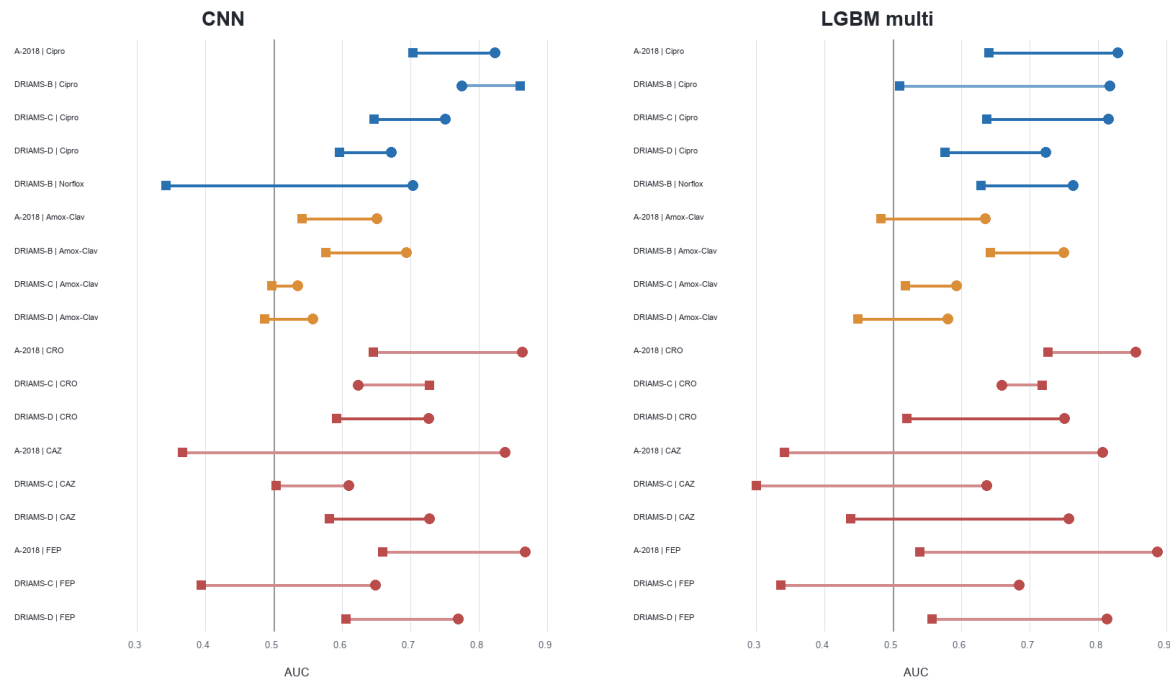
Preview shown above; full table is saved as table_1_primary_background_matched_audit.csv.

Figure 1. Raw external AUC versus background-centered AUC

Raw external AUC vs background-centered AUC

Circle = raw external AUC; square = after matching/centering by co-resistance background.

fluoroquinolone cephalosporin/ESBL mixed beta-lactam



What it measures. Each line connects raw external AUC to background-centered AUC for the same model, site, and drug. The circle is raw AUC; the square is the AUC after controlling for co-resistance background.

What a reviewer should see. The same raw performance can mean different things after background control. Some pairs retain meaningful centered AUC, while others move toward chance or have low matched support.

How it supports the paper. This is the cleanest visual form of the argument. If raw AUC were sufficient, the squares would track the circles closely across drugs. Instead, the distance between them reveals background sensitivity.

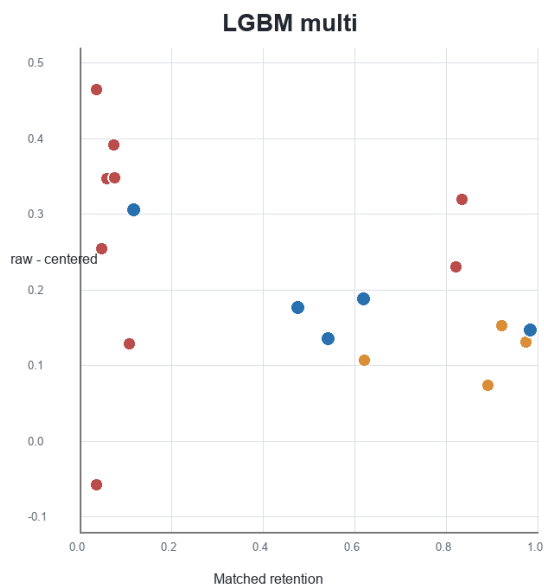
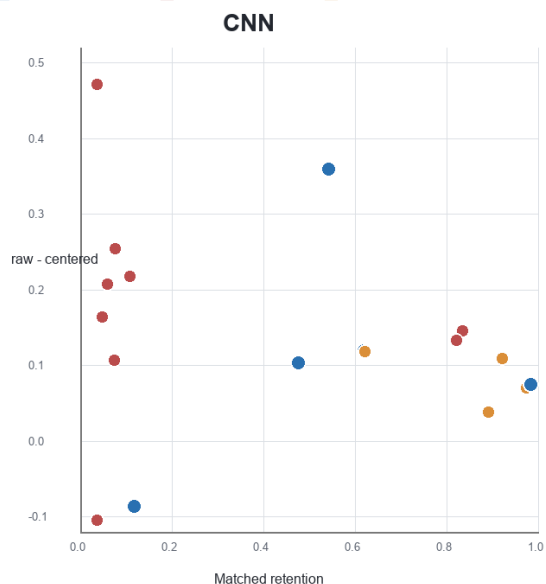
Caveat. Background-centered AUC is not causal proof of mechanism. It is an audit statistic showing whether prediction survives a specific observable background control.

Figure 2. Signal drop versus matched retention

Background sensitivity: signal drop vs matched retention

Large positive drop means raw AUC was higher than background-centered AUC.

fluoroquinolone cephalosporin/ESBL mixed beta-lactam



What it measures. The x-axis is the fraction of examples retained after background matching. The y-axis is raw AUC minus background-centered AUC. High y-values indicate a large apparent loss after controlling background.

What a reviewer should see. This figure separates real collapses from unreliable estimates. A point with large drop and high retention is concerning; a point with low retention is treated as cautionary.

How it supports the paper. It prevents overclaiming. The method does not simply call every post-matching drop biologically meaningful. It asks whether there were enough matched strata to interpret the result.

Caveat. Sparse resistance patterns can make matching impossible for some drug/site combinations. Those cases support the need for the audit, but they should not be used as strong pair-level conclusions.

Table 2. CNN versus LGBM multi background audit

What it measures. This table repeats the audit across model families: the CNN/Mega model and the LGBM multi-task baseline.

Key result. Background sensitivity appears in both model families. Amox-Clav is repeatedly weak or collapses after matching, while Cipro retains more residual signal.

How it supports the paper. This blocks the simplest reviewer objection: that the result is just a neural-network artifact. The audit is about MALDI-AMR prediction under structured labels, not only one architecture.

Reviewer caveat. The locally verified final artifact includes CNN and LGBM multi, not LGBM single. LGBM single should be added if we want the strongest possible model-family replication figure.

site	drug	cnn_raw_auc	cnn_centered_auc	cnn_raw_minus_centered	lgbm_raw_auc	lgbm_centered_auc	lgbm_raw_minus_centered	model_family_consensus
A-2018	Amox-Clav	0.650	0.541	0.109	0.635	0.482	0.153	collapses after matching
A-2018	FEP	0.867	0.659	0.208	0.886	0.539	0.347	model-dependent / mixed
A-2018	CAZ	0.838	0.367	0.471	0.806	0.342	0.465	low support
A-2018	CRO	0.863	0.645	0.218	0.854	0.726	0.129	survives background matching
A-2018	Cipro	0.823	0.703	0.120	0.828	0.640	0.188	survives background matching
A-2018	Norflox	0.743			0.744			not matched
DRIAMS-B	Amox-Clav	0.694	0.576	0.118	0.749	0.642	0.107	residual signal in both models
DRIAMS-B	FEP	0.753			0.844			not matched
DRIAMS-B	CAZ	0.735			0.802			not matched
DRIAMS-B	CRO	0.743			0.816			not matched
DRIAMS-B	Cipro	0.774	0.860	-0.086	0.816	0.510	0.306	model-dependent / mixed
DRIAMS-B	Norflox	0.703	0.343	0.360	0.763	0.628	0.136	model-dependent / mixed

Preview shown above; full table is saved as `table_2_cnn_vs_lgbm_multi_background_audit.csv`.

Table 3. Ecology interpretation table

What it measures. This table attaches each audit result to its resistance ecology block and strongest co-resistance partner.

Key result. The drugs are not independent labels. Fluoroquinolones and cephalosporins form strong ecology blocks, and the model can exploit these blocks.

How it supports the paper. It connects the statistical audit to biology. The model is not choosing from isolated drug labels; it sees spectra from bacteria embedded in co-resistance structure.

Reviewer caveat. Ecology blocks are inferred from AST co-resistance patterns, not from direct WGS labels in DRIAMS. They are a measurable proxy, not final phylogenetic proof.

site	drug	resistance_ecology_block	strongest_network_partner	partner_phi	cnn_drop	lgbm_drop	interpretation
A-2018	Amox-Clav	mixed beta-lactam/background	CAZ	0.379	0.109	0.153	background-driven in both model families
A-2018	FEP	cephalosporin/E SBL block	CAZ	0.828	0.208	0.347	partial residual focal signal
A-2018	CAZ	cephalosporin/E SBL block	CRO	0.884	0.471	0.465	background-driven in both model families

site	drug	resistance_ecology_block	strongest_network_partner	partner_phi	cnn_drop	lgbm_drop	interpretation
A-2018	CRO	cephalosporin/E SBL block	CAZ	0.884	0.218	0.129	focal signal survives background matching
A-2018	Cipro	fluoroquinolone block	Norflox	0.976	0.120	0.188	focal signal survives background matching
A-2018	Norflox	fluoroquinolone block	Cipro	0.976			insufficient matched support
DRIAMS-B	Amox-Clav	mixed beta-lactam/background	CAZ	0.379	0.118	0.107	partial residual focal signal
DRIAMS-B	FEP	cephalosporin/E SBL block	CAZ	0.828			insufficient matched support
DRIAMS-B	CAZ	cephalosporin/E SBL block	CRO	0.884			insufficient matched support
DRIAMS-B	CRO	cephalosporin/E SBL block	CAZ	0.884			insufficient matched support
DRIAMS-B	Cipro	fluoroquinolone block	Norflox	0.976	-0.086	0.306	partial residual focal signal
DRIAMS-B	Norflox	fluoroquinolone block	Cipro	0.976	0.360	0.136	partial residual focal signal

Preview shown above; full table is saved as `table_3_ecology_interpretation.csv`.

Table 4. Locked transfer prediction assessment

What it measures. This table compares pre-specified transfer predictions against observed external behavior in the E. coli six-drug panel.

Key result. The simple transferability prediction was partially right but not perfect: Cipro/Norfloxacin transfer more than Amox-Clav, but cephalosporins show hospital-dependent behavior.

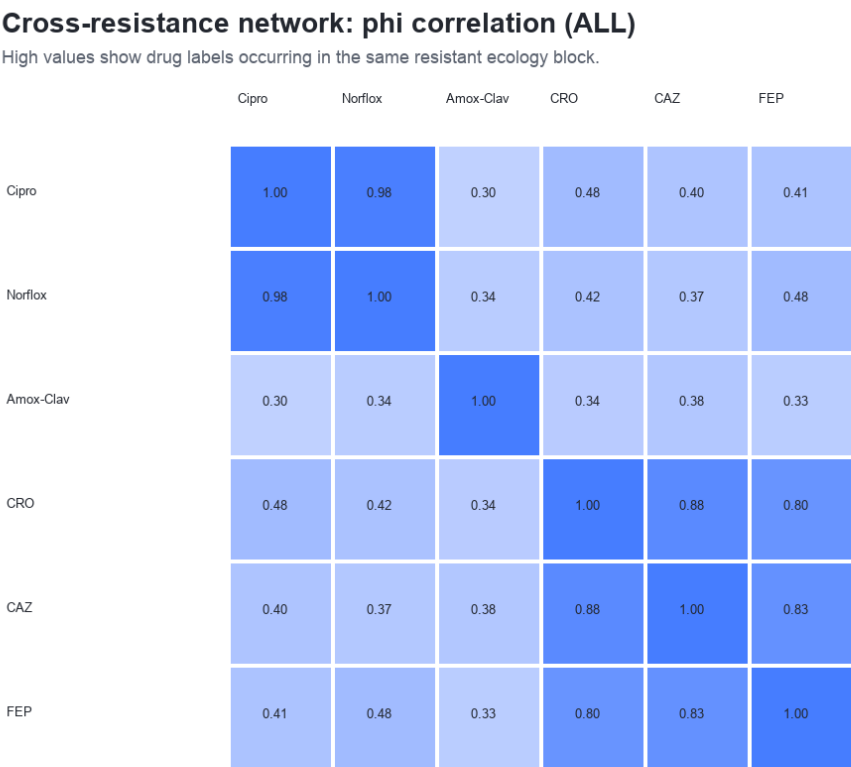
How it supports the paper. The table moves the story away from a naive mechanism-only hypothesis and toward population/hospital ecology. It shows why the background audit is necessary.

Reviewer caveat. This should not be framed as a perfect prospective score. It is better used as evidence that source AUC alone is insufficient and hospital-specific resistant ecology matters.

drug	block	locked_prediction	observed_external_category	prediction_assessment	auc_A-2018	auc_DRIAMS-B	auc_DRIAMS-C	auc_DRIAMS-D
CAZ	ESBL/AmpC cephalosporin block	weaker_transfer	partial_transfer	partially_confirmed	0.843	0.771	0.628	0.748
CRO	ESBL/AmpC cephalosporin block	weaker_transfer	partial_transfer	partially_confirmed	0.873	0.776	0.644	0.744
FEP	ESBL/AmpC cephalosporin block	weaker_transfer	transfer	contradicted_by_mean_auc	0.874	0.815	0.670	0.798
Cipro	Fluoroquinolone block	transfer	partial_transfer	partially_confirmed	0.833	0.794	0.760	0.680
Norflox	Fluoroquinolone block	transfer	transfer	confirmed	0.767	0.733	0.782	
Amox-Clav	Heterogeneous inhibitor beta-lactam	partial_weak	weak_transfer	confirmed	0.650	0.724	0.542	0.556

Preview shown above; full table is saved as `table_4_transfer_prediction_assessment.csv`.

Figure 3. Cross-resistance phi heatmap



What it measures. The heatmap shows phi correlations between drug-resistance labels across the E. coli panel. High values mean two resistances tend to appear in the same isolates.

What a reviewer should see. Cipro/Norfloxacin form a strong fluoroquinolone block. Ceftriaxone, Ceftazidime, and Cefepime form a strong cephalosporin/ESBL block.

How it supports the paper. This explains why a model can appear to learn a focal drug while actually leveraging background. If drug labels co-occur, the spectral identity of a resistant subpopulation can help predict multiple drugs.

Caveat. Co-resistance correlation is not equivalent to clonal identity. It shows exploitable label ecology, not the exact evolutionary history.

Table 5. Top cross-resistance network edges

What it measures. This table ranks the strongest pairwise co-resistance relationships by phi correlation and related measures.

Key result. The strongest edges are exactly the expected ecology blocks: Cipro/Norfloxacin and the cephalosporin cluster.

How it supports the paper. It supplies the label-structure evidence behind background matching. The audit is not controlling for an imaginary confound; the confound is visible in the AST matrix.

Reviewer caveat. The strongest edges depend on which drugs are assayed at each site and how missing AST calls are handled.

site	drug_a	drug_b	n_both_known	n_rr	rr_lift	phi	resistant_jaccard
ALL	Cipro	Norflox	1090	263	3.993	0.976	0.963
ALL	CRO	CAZ	4349	446	7.841	0.884	0.811
ALL	CAZ	FEP	3907	271	10.212	0.828	0.719

site	drug_a	drug_b	n_both_known	n_rr	rr_lift	phi	resistant_jaccard
ALL	CRO	FEP	3956	283	9.394	0.804	0.680
ALL	Norflox	FEP	771	89	2.908	0.479	0.392
ALL	Cipro	CRO	4371	437	3.223	0.479	0.382
ALL	Norflox	CRO	1079	123	2.559	0.419	0.362
ALL	Cipro	FEP	3873	222	3.988	0.414	0.273
ALL	Cipro	CAZ	4246	310	3.263	0.402	0.302
ALL	Amox-Clav	CAZ	4304	318	3.014	0.379	0.287
ALL	Norflox	CAZ	1068	101	2.545	0.372	0.314
ALL	Amox-Clav	CRO	4439	369	2.540	0.342	0.289

Preview shown above; full table is saved as `table_5_top_cross_resistance_edges.csv`.

Table 6. E. coli block-site summary

What it measures. This table compresses the E. coli six-drug panel into ecological blocks and summarizes source-to-external behavior.

Key result. Block-level external behavior varies by site. The cephalosporin block can look strong in one hospital and weaker in another, reinforcing that transferability is not a fixed drug property.

How it supports the paper. It supports the shift from mechanism-only framing to hospital/ecology-aware evaluation.

Reviewer caveat. A block mean can hide within-block heterogeneity; use it as a summary, not a replacement for pair-level audit results.

site	block	mean_auc	mean_aupr	n_pairs	n	n_r	auc_source_delta
A-2018	ESBL/AmpC cephalosporin block	0.863	0.562	3	3877	443	0.000
A-2018	Fluoroquinolone block	0.800	0.620	2	1789	442	0.000
A-2018	Heterogeneous inhibitor beta-lactam	0.650	0.442	1	1367	383	0.000
DRIAMS-B	ESBL/AmpC cephalosporin block	0.787	0.533	3	639	133	-0.076
DRIAMS-B	Fluoroquinolone block	0.763	0.640	2	422	127	-0.036
DRIAMS-B	Heterogeneous inhibitor beta-lactam	0.724	0.534	1	198	61	0.073
DRIAMS-C	ESBL/AmpC cephalosporin block	0.647	0.311	3	2430	400	-0.216
DRIAMS-C	Fluoroquinolone block	0.771	0.634	2	1328	306	-0.029
DRIAMS-C	Heterogeneous inhibitor beta-lactam	0.542	0.302	1	902	230	-0.108
DRIAMS-D	ESBL/AmpC cephalosporin block	0.763	0.361	3	5904	456	-0.100
DRIAMS-D	Fluoroquinolone block	0.680	0.390	1	1939	371	-0.119
DRIAMS-D	Heterogeneous inhibitor beta-lactam	0.556	0.255	1	1994	332	-0.094

Preview shown above; full table is saved as `table_6_ecoli_block_site_summary.csv`.

Figure 4. Public WGS-linked MALDI and proteomic support

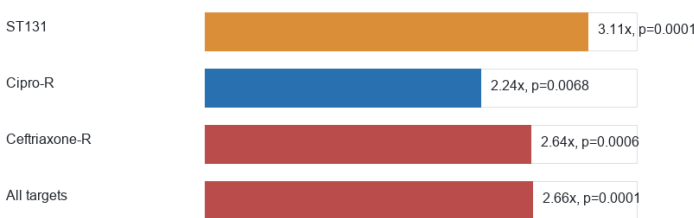
Public WGS-linked MALDI support

Lineage is strongly encoded in Bruker MALDI peaks; resistance-associated peaks overlap published ST131 markers.

A. MALDI peak AUC in public Basel UPEC dataset



B. Enrichment for published ST131 MALDI biomarkers



What it measures. Panel A summarizes MALDI peak-based discrimination of ST131, Cipro resistance, and Ceftriaxone resistance in a public WGS-linked Basel UPEC dataset. Panel B summarizes enrichment of discriminative peaks for published ST131 MALDI biomarkers.

What a reviewer should see. ST131 is more strongly encoded in MALDI spectra than either resistance phenotype. Resistance-associated peaks are non-randomly enriched for published ST131 biomarkers.

How it supports the paper. This is the biological bridge. It supports the idea that MALDI spectra can carry lineage signal strongly enough to contaminate or support apparent AMR prediction.

Caveat. This is not direct DRIAMS WGS validation and not direct MS/MS protein identification. It is public WGS-linked and literature-backed support for the plausibility of the background signal.

Table 7. Public WGS-linked MALDI AUCs

What it measures. This table reports peak-based discrimination in the public WGS-linked UPEC dataset.

Key result. ST131 AUC is 0.906, higher than Cipro-R and Ceftriaxone-R.

How it supports the paper. It establishes that lineage can be a very strong MALDI signal in an independent public dataset.

Reviewer caveat. The classifier is a simple centroid-direction model, useful as a transparent signal check rather than a final optimized predictor.

target	n	class_0	class_1	auc	folds	model
ST131	407	352	55	0.906	5	centroid_direction
Ciprofloxacin_R	350	297	53	0.739	5	centroid_direction
Ceftriaxone_R	360	330	30	0.650	5	centroid_direction

Preview shown above; full table is saved as `table_7_public_wgs_maldi_auc.csv`.

Table 8. ST131 resistance associations

What it measures. This table shows how strongly resistance phenotypes are associated with ST131 in the public WGS-linked dataset.

Key result. Both Cipro and Ceftriaxone resistance are strongly enriched in ST131 relative to non-ST131 isolates.

How it supports the paper. It explains why MALDI lineage signal could be useful for apparent AMR prediction: resistance is not randomly distributed across lineages.

Reviewer caveat. Association is not causality. This table supports confounding plausibility, not direct mechanism.

phenotype	st131_resistant	st131_susceptible	nonst131_resistant	nonst131_susceptible	haldane_odds_ratio
Ciprofloxacin	30	17	23	280	20.803
Ceftriaxone	21	27	9	303	24.977

Preview shown above; full table is saved as `table_8_public_wgs_st131_resistance_associations.csv`.

Table 9. Published ST131 biomarker enrichment

What it measures. This table compares observed overlap between discriminative MALDI bins and published ST131 marker peaks against mass-matched random permutations.

Key result. The overlap is non-random for ST131 and remains enriched for Cipro-R and Ceftriaxone-R discriminative peaks.

How it supports the paper. It gives proteomic grounding to the lineage-background argument. Resistance-associated peaks are not arbitrary; they overlap known ST131 MALDI biomarkers more than expected by chance.

Reviewer caveat. This is biomarker overlap, not direct protein identification from our spectra. Use language like 'enriched for published markers,' not 'identified as protein X.'

target	observed_overlap_count	top_peak_count	null_mean_overlap_count	fold_enrichment	z_score	empirical_p_ge_observed
ST131	14	75	4.506	3.107	5.128	0.000
Ciprofloxacin_R	10	75	4.474	2.235	2.971	0.007
Ceftriaxone_R	12	75	4.539	2.644	3.995	0.001
ALL_TARGETS	36	225	13.550	2.657	8.739	0.000

Preview shown above; full table is saved as `table_9_published_st131_biomarker_enrichment.csv`.

Table 10. Top published ST131 peak overlaps

What it measures. This table lists the closest individual mass overlaps between discriminative MALDI bins and published ST131 biomarker masses.

Key result. Several top ST131/resistance-associated bins align with published ST131 markers such as HdeA/YjbJ-related masses.

How it supports the paper. It makes the enrichment result interpretable by showing concrete masses and candidate annotations.

Reviewer caveat. Mass overlap alone is not definitive protein assignment. This table is hypothesis-generating and supportive, not final proteomics proof.

target	mz_center	published_mz	delta_da	protein	annotation
Ceftriaxone_R	11787.500	11783	4.500	cytochrome b562	soluble cytochrome b562
Ceftriaxone_R	4862.500	4857	5.500	HdeA multivalent ion	multivalent ion of HdeA m/z 9710
Ceftriaxone_R	8437.500	8448	10.500	YnfD	uncharacterized protein YnfD
Ceftriaxone_R	6837.500	6827	10.500	unidentified	ST131-specific peak not identified by LC-MS/MS
Ceftriaxone_R	8337.500	8351	13.500	YjbJ	UPF0337 protein YjbJ

target	mz_center	published_mz	delta_da	protein	annotation
Ciprofloxacin_R	11787.500	11783	4.500	cytochrome b562	soluble cytochrome b562
Ciprofloxacin_R	4862.500	4857	5.500	HdeA multivalent ion	multivalent ion of HdeA m/z 9710
Ciprofloxacin_R	7662.500	7655	7.500	YahO	uncharacterized protein YahO
Ciprofloxacin_R	8437.500	8448	10.500	YnfD	uncharacterized protein YnfD
Ciprofloxacin_R	6837.500	6827	10.500	unidentified	ST131-specific peak not identified by LC-MS/MS
ST131	9712.500	9710	2.500	HdeA	acid stress chaperone HdeA
ST131	11787.500	11783	4.500	cytochrome b562	soluble cytochrome b562

Preview shown above; full table is saved as `table_10_top_published_st131_peak_overlaps.csv`.

Integrated Argument

The argument is cumulative. Table 1 and Figure 1 show that raw external performance changes after background matching. Figure 2 shows that the framework distinguishes interpretable collapses from low-support matched strata. Table 2 shows the phenomenon is not restricted to one neural model. Tables 3, 5, and Figure 3 show why the phenomenon is plausible: resistance labels are structured into co-resistance ecology blocks. Tables 7 to 10 and Figure 4 then show that lineage signal is strongly encoded in public WGS-linked MALDI data and overlaps known ST131 biomarkers.

The resulting claim is conservative but important: MALDI-AMR models can learn the spectral distribution of resistant bacteria as they occur in hospital populations. Therefore, raw AUC alone is insufficient for interpreting whether a model has learned focal-drug resistance biology.

- Defensible claim: background-matched evaluation is needed before interpreting MALDI-AMR predictions as focal resistance detection.
- Defensible claim: the current data show background sensitivity across CNN and LGBM multi models.
- Defensible claim: public WGS-linked MALDI data show lineage, especially ST131, is strongly encoded in spectra.
- Do not claim yet: exact protein identities for DRIAMS peaks.
- Do not claim yet: DRIAMS models directly detect ST131 without DRIAMS WGS/MLST labels.

What This Would Need For A Higher-Tier Submission

The framework is already coherent. The main missing pieces for a stronger Nature Communications-style case are: LGBM single or logistic regression audit in the same final artifact pack, WGS-linked clone-control analysis in the public UPEC dataset, and a cleaner software release with a one-command audit workflow.

The most valuable next analysis is not more raw DRIAMS AUC. It is a WGS-linked clone-control audit: test whether Cipro/Ceftriaxone resistance prediction remains after controlling ST131 or sequence type in the public UPEC dataset.